



# Transitional Cell Cancer of the Renal Pelvis and Ureter Treatment (PDQ®)–Patient Version

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## General Information About Transitional Cell Cancer of the Renal Pelvis and Ureter

### KEY POINTS

- Transitional cell cancer of the renal pelvis and ureter is a type of cancer that forms in the transitional cells of the renal pelvis and ureter.
- A personal history of bladder cancer and smoking can increase the risk of transitional cell cancer of the renal pelvis and ureter.
- Signs and symptoms of transitional cell cancer of the renal pelvis and ureter include blood in the urine and back pain.
- Tests that examine the abdomen and kidneys are used to diagnose transitional cell cancer of the renal pelvis and ureter.
- After transitional cell cancer of the renal pelvis and ureter has been diagnosed, tests are done to find out if cancer cells have spread within the renal pelvis and ureter or to other parts of the body.
- Some people decide to get a second opinion.
- Certain factors affect prognosis (chance of recovery) and treatment options.

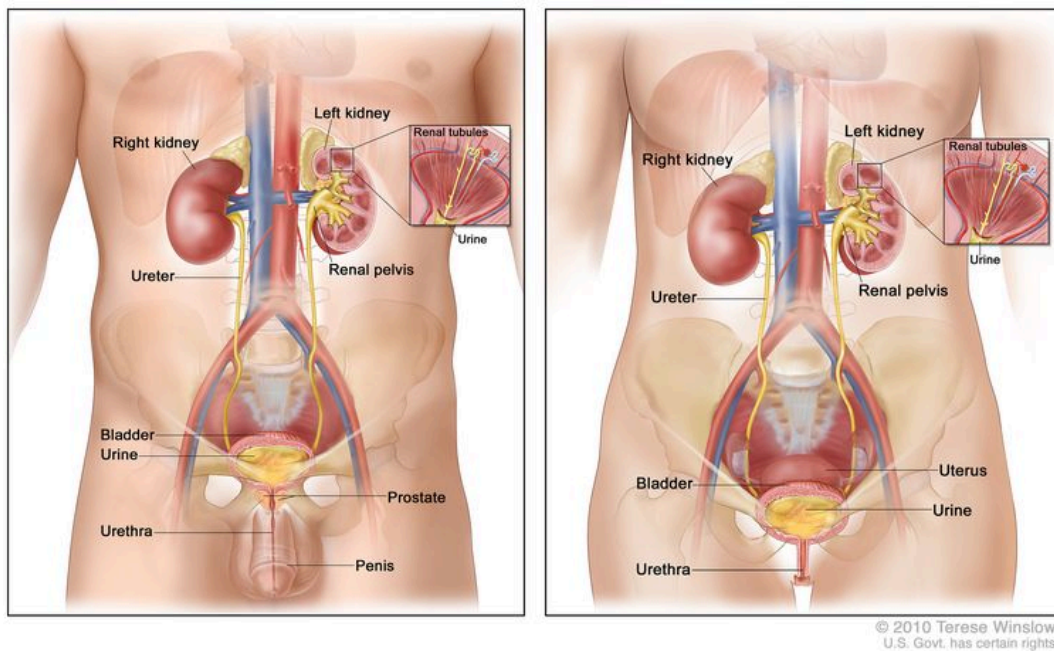
**Transitional cell cancer of the renal pelvis and ureter is a type of cancer that forms in the transitional cells of the renal pelvis and ureter.**

Transitional cells are a type of cell that lines the inside of the renal pelvis, ureters, and other organs. These cells are able to stretch when the renal pelvis or bladder is full of urine and shrink when it is emptied. Transitional cell cancer can form in the renal pelvis, the ureters, or both.

The renal pelvis is the area at the center of the kidney. It collects urine before it flows into the ureters, which are long tubes that connect the kidney to the bladder. There are two kidneys, one on each side of the backbone, above the waist. The kidneys of an adult are about 5 inches long and 3 inches wide and are shaped like a kidney bean.

The kidneys play an important role in the urinary system:

- The kidneys filter and clean the blood by taking out waste products and extra water as urine.
- The urine collects in the middle of each kidney in the renal pelvis.
- Urine passes from the renal pelvis through the ureter into the bladder.
- The bladder holds the urine until it passes through the urethra and leaves the body.



Anatomy of the male urinary system (left panel) and female urinary system (right panel) showing the kidneys, ureters, bladder, and urethra. The inside of the left kidney shows the renal pelvis. An inset shows the renal tubules and urine. Also shown are the prostate and penis (left panel) and the uterus (right panel). Urine is made in the renal tubules and collects in the renal pelvis of each kidney. The urine flows from the kidneys through the ureters to the bladder. The urine is stored in the bladder until it leaves the body through the urethra.

Renal cell cancer is a more common type of kidney cancer. For more information, see [Renal Cell Cancer Treatment](#).

## **A personal history of bladder cancer and smoking can increase the risk of transitional cell cancer of the renal pelvis and ureter.**

Transitional cell cancer of the renal pelvis and ureter is caused by certain changes to the way transitional cells in the renal pelvis and ureters function, especially how they grow and divide into new cells. There are many risk factors for transitional cell cancer of the renal pelvis and ureter, but many do not directly cause cancer. Instead, they increase the chance of DNA damage in cells that may lead to cancer. To learn more about how cancer develops, see [What Is Cancer?](#)

A risk factor is anything that increases the chance of getting a disease. Some risk factors for transitional cell cancer of the renal pelvis and ureter, like smoking, can be changed. However, risk factors may also include things you cannot change, like your personal medical history. Learning about risk factors for transitional cell cancer of the renal pelvis and ureter can help you make changes that might lower your risk of getting it.

Risk factors for transitional cell cancer of the renal pelvis and ureter include:

- having a personal history of bladder cancer
- smoking cigarettes
- taking a lot of certain pain medicines, such as phenacetin
- being exposed to certain dyes and chemicals used in making leather goods, textiles, plastics, and rubber

Having one or more of these risk factors does not mean that you will get transitional cell cancer of the renal pelvis and ureter. Many people with risk factors never develop this type of cancer, while others with no known risk factors do. Talk with your doctor if you think you might be at risk.

## **Signs and symptoms of transitional cell cancer of the renal pelvis and ureter include blood in the urine and back pain.**

In the early stages, there may be no signs and symptoms of transitional cell cancer of the renal pelvis and ureter. Symptoms may appear as the tumor grows and may include:

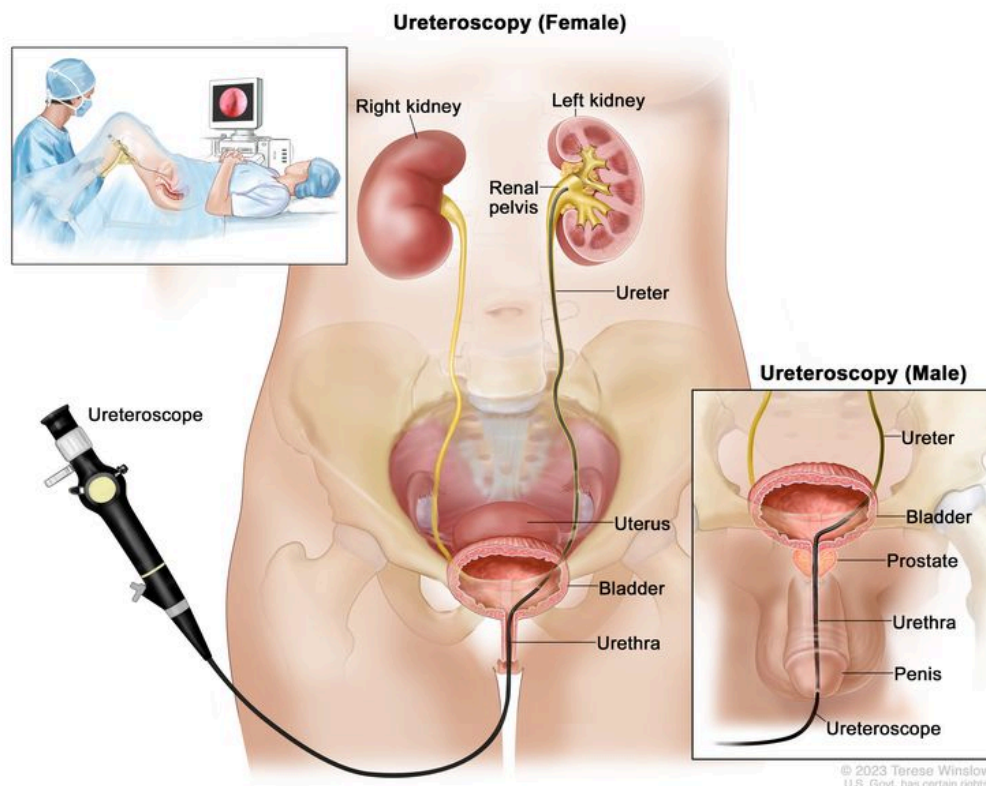
- blood in the urine
- a pain in the back that doesn't go away
- extreme tiredness
- weight loss with no known reason
- painful or frequent urination

These symptoms may be caused by transitional cell cancer of the renal pelvis and ureter or by other conditions. It's important to check with your doctor if you have any of these symptoms to find out the cause and begin treatment if needed.

## Tests that examine the abdomen and kidneys are used to diagnose transitional cell cancer of the renal pelvis and ureter.

In addition to asking about your personal and family health history and doing a physical exam, your doctor may perform the following tests and procedures:

- **Urinalysis** is a test to check the color of urine and its contents, such as sugar, protein, blood, and bacteria.
- **Ureteroscopy** is a procedure to look inside the ureter and renal pelvis to check for abnormal areas. A ureteroscope is a thin, tube-like instrument with a light and a lens for viewing. The ureteroscope is inserted through the urethra into the bladder, ureter, and renal pelvis. A tool may be inserted through the ureteroscope to take tissue samples to be checked under a microscope for signs of disease.



A ureteroscopy is a procedure that uses a ureteroscope (a thin, tube-like instrument with a light and a lens for viewing) to look inside the ureter and renal pelvis to check for abnormal areas. The ureteroscope is inserted through the urethra into the bladder, ureter, and renal pelvis.

- **Urine cytology** is a laboratory test in which a sample of urine is checked under a microscope for abnormal cells. Cancer in the kidney, bladder, or ureter may shed cancer cells into the urine.
- **CT scan (CAT scan)** uses a computer linked to an x-ray machine to make a series of detailed pictures of areas inside the body from different angles. A dye may be injected into a vein or swallowed to help the organs or tissues show up more clearly. This procedure is also called computed tomography, computerized tomography, or computerized axial tomography.
- **Ultrasound** uses high-energy sound waves (ultrasound) that bounce off internal tissues or organs and make echoes. The echoes form a picture of body tissues called a sonogram. An ultrasound of the abdomen may be done to help diagnose cancer of the renal pelvis and ureter.
- **MRI (magnetic resonance imaging)** uses a magnet, radio waves, and a computer to make a series of detailed pictures of areas inside the body, such as the pelvis. This procedure is also called nuclear magnetic resonance imaging (NMRI).
- **Biopsy** is the removal of cells or tissues so they can be viewed under a microscope by a pathologist to check for signs of cancer. This may be done during a ureteroscopy or surgery.

**After transitional cell cancer of the renal pelvis and ureter has been diagnosed, tests are done to find out if cancer cells have spread within the renal pelvis and ureter or to other parts of the body.**

The process used to find out if the cancer has spread within the renal pelvis and ureter or to other parts of the body is called staging. The information gathered from the staging process determines the stage of the disease. The results of the diagnostic and staging tests will also help you and your doctor plan treatment.

The following tests and procedures may also be used in the staging process:

- **Chest x-ray** is an x-ray of the organs and bones inside the chest. An x-ray is a type of energy beam that can go through the body and onto film, making a picture of areas inside the body.
- **PET scan (positron emission tomography scan)** uses a small amount of radioactive sugar (also called glucose) that is injected into a vein. Then a scanner is used to make detailed, computerized pictures of areas inside the body where the glucose is taken up. Because cancer cells often take up more glucose than normal cells, the pictures can be used to find cancer cells in the body.

- **Bone scan** is used to check if there are rapidly dividing cells, such as cancer cells, in the bone. A very small amount of radioactive material is injected into a vein and travels through the bloodstream. The radioactive material collects in the bones with cancer and is detected by a scanner.

## Some people decide to get a second opinion.

You may want to get a second opinion to confirm your cancer diagnosis and treatment plan. If you seek a second opinion, you will need to get medical test results and reports from the first doctor to share with the second doctor. The second doctor will review the pathology report, slides, and scans. They may agree with the first doctor, suggest changes or another treatment approach, or provide more information about your cancer.

To learn more about choosing a doctor and getting a second opinion, see [Finding Cancer Care](#). You can contact [NCI's Cancer Information Service](#) via chat, email, or phone (both in English and Spanish) for help finding a doctor, hospital, or getting a second opinion. For questions you might want to ask at your appointments, see [Questions to Ask Your Doctor about Cancer](#).

## Certain factors affect prognosis (chance of recovery) and treatment options.

The prognosis depends on the stage and grade of the tumor.

The treatment options depend on:

- the stage and grade of the tumor
- where the tumor is
- whether the patient's other kidney is healthy
- whether the cancer has recurred (come back)

Most transitional cell cancer of the renal pelvis and ureter can be cured if found early.

## Stages of Transitional Cell Cancer of the Renal Pelvis and Ureter

### KEY POINTS

- The following stages are used for transitional cell cancer of the renal pelvis and/or ureter:
  - Stage 0 (Noninvasive Papillary Carcinoma and Carcinoma in Situ)



- Stage I (also called stage 1)
- Stage II (also called stage 2)
- Stage III (also called stage 3)
- Stage IV (also called stage 4)
- Transitional cell cancer of the renal pelvis and ureter is also described as localized, regional, metastatic, or recurrent:
  - Localized
  - Regional
  - Metastatic
  - Recurrent

Cancer stage describes the extent of cancer in the body, such as the size of the tumor, whether it has spread, and how far it has spread from where it first formed. It is important to know the stage of transitional cell cancer of the renal pelvis and ureter to plan the best treatment.

There are several staging systems for cancer that describe the extent of the cancer. Transitional cell cancer of the renal pelvis and ureter staging usually uses the TNM staging system. You may see your cancer described by this staging system in your pathology report. Based on the TNM results, a stage (I, II, III, or IV, also written as 1, 2, 3, or 4) is assigned to your cancer. When talking to you about your cancer, your doctor may describe it as one of these stages.

Learn more about [Cancer Staging](#).

## **The following stages are used for transitional cell cancer of the renal pelvis and/or ureter:**

### **Stage 0 (Noninvasive Papillary Carcinoma and Carcinoma in Situ)**

In stage 0, abnormal cells are found in tissue lining the inside of the renal pelvis or ureter. These abnormal cells may become cancer and spread into nearby normal tissue. Stage 0 is divided into stages 0a and 0is, depending on the type of tumor:

- Stage 0a is also called noninvasive papillary carcinoma, which may look like long, thin growths that grow out from the tissue lining the inside of the renal pelvis or ureter.
- Stage 0is is also called carcinoma in situ, which is a flat tumor on the tissue lining the inside of the renal pelvis or ureter.

## **Stage I (also called stage 1)**

In stage I, cancer has formed and has spread from the tissue lining the inside of the renal pelvis or ureter to the connective tissue layer.

## **Stage II (also called stage 2)**

In stage II, cancer has spread to the muscle layer of the renal pelvis or ureter.

## **Stage III (also called stage 3)**

In stage III, cancer has spread:

- from the muscle layer of the renal pelvis to fat around the renal pelvis or to tissue in the kidney; or
- from the muscle layer of the ureter to fat around the ureter.

## **Stage IV (also called stage 4)**

In stage IV, cancer has spread to at least one of the following:

- a nearby organ.
- lymph nodes.
- other parts of the body, such as the lung, liver, or bone.

Stage IV transitional cell cancer of the renal pelvis and ureter is also called metastatic cancer. Metastatic cancer happens when cancer cells travel through the lymphatic system or blood and form tumors in other parts of the body. The metastatic tumor is the same type of cancer as the primary tumor. For example, if transitional cell cancer renal pelvis and ureter spreads to the liver, the cancer cells in the liver are actually transitional cell cancer of the renal pelvis and ureter. The disease is called metastatic transitional cell cancer of the renal pelvis and ureter, not liver cancer. Learn more in [Metastatic Cancer: When Cancer Spreads](#).

## **Transitional cell cancer of the renal pelvis and ureter is also described as localized, regional, metastatic, or recurrent:**

### **Localized**

The cancer is found only in the kidney.

### **Regional**

The cancer has spread to tissues around the kidney and to nearby lymph nodes and blood vessels in the pelvis.



## Metastatic

The cancer has spread to other parts of the body.

## Recurrent

Recurrent cancer is cancer that has recurred (come back) after it has been treated. If transitional cell cancer of the renal pelvis and ureter comes back, it may come back in the renal pelvis, ureter, or in other parts of the body, such as the lung, liver, or bone. Tests will be done to help determine where the cancer has returned. The type of treatment for recurrent transitional cell cancer of the renal pelvis and ureter will depend on where it has come back.

Learn more in [Recurrent Cancer: When Cancer Comes Back](#).

# Treatment Option Overview for Transitional Cell Cancer of the Renal Pelvis and Ureter

## KEY POINTS

- There are different types of treatment for patients with transitional cell cancer of the renal pelvis and ureter.
- The following type of treatment is used:
  - Surgery
- New types of treatment are being tested in clinical trials.
  - Fulguration
  - Segmental resection of the renal pelvis
- Treatment for transitional cell cancer of the renal pelvis and ureter may cause side effects.
- Follow-up care may be needed.

## There are different types of treatment for patients with transitional cell cancer of the renal pelvis and ureter.

Different types of treatments are available for people with transitional cell cancer of the renal pelvis and ureter. You and your cancer care team will work together to decide your treatment plan, which may include more than one type of treatment. Many factors will be considered, such as the stage of the cancer, your overall health, and your preferences. Your plan will

include information about your cancer, the goals of treatment, your treatment options and the possible side effects, and the expected length of treatment.

Talking with your cancer care team before treatment begins about what to expect will be helpful. You'll want to learn what you need to do before treatment begins, how you'll feel while going through it, and what kind of help you will need. To learn more, see [Questions to Ask Your Doctor about Treatment](#).

## **The following type of treatment is used:**

### **Surgery**

One of the following surgical procedures may be used to treat transitional cell cancer of the renal pelvis and ureter:

- Nephroureterectomy is surgery to remove the entire kidney, the ureter, and the bladder cuff (tissue that connects the ureter to the bladder). Most people with transitional cell cancer of the renal pelvis and ureter undergo nephroureterectomy.
- Segmental resection of the ureter is a surgical procedure to remove the part of the ureter that contains cancer and some of the healthy tissue around it. The ends of the ureter are then reattached. This treatment is used when the cancer is superficial and in the lower third of the ureter only, near the bladder.

## **New types of treatment are being tested in clinical trials.**

For some people, joining a clinical trial may be an option. There are different types of clinical trials for people with cancer. For example, a treatment trial tests new treatments or new ways of using current treatments. Supportive care and palliative care trials look at ways to improve quality of life, especially for those who have side effects from cancer and its treatment.

You can use the [clinical trial search](#) to find NCI-supported cancer clinical trials accepting participants. The search allows you to filter trials based on the type of cancer, your age, and where the trials are being done. Clinical trials supported by other organizations can be found on the [ClinicalTrials.gov](#) website.

Learn more about clinical trials, including how to find and join one, at [Clinical Trials Information for Patients and Caregivers](#).

This summary section describes treatments that are being studied in clinical trials. It may not mention every new treatment being studied.

### **Fulguration**

Fulguration is a surgical procedure that destroys tissue using an electric current. A tool with a small wire loop on the end is used to remove the cancer or to burn away the tumor with electricity.

## **Segmental resection of the renal pelvis**

This is a surgical procedure to remove localized cancer from the renal pelvis without removing the entire kidney. Segmental resection may be done to save kidney function when the other kidney is damaged or has already been removed.

## **Laser surgery**

A laser beam (narrow beam of intense light) is used as a knife to remove the cancer. A laser beam can also be used to kill the cancer cells. This procedure may also be called laser fulguration.

## **Regional chemotherapy and regional immunotherapy**

Chemotherapy (also called chemo) uses drugs to stop the growth of cancer cells, either by killing the cells or by stopping the cells from dividing. To learn more about chemotherapy, see [Chemotherapy to Treat Cancer](#) and [Chemotherapy and You: Support for People With Cancer](#).

Immunotherapy helps a person's immune system fight cancer. Learn more about [Immunotherapy to Treat Cancer](#).

Regional treatment for transitional cell cancer of renal pelvis and ureter is a method of placing the chemotherapy or immunotherapy drug directly into the renal pelvis or the ureter to mainly affect cancer cells in those areas.

## **Treatment for transitional cell cancer of the renal pelvis and ureter may cause side effects.**

For information about side effects caused by treatment for cancer, visit our [Side Effects](#) page.

## **Follow-up care may be needed.**

As you go through treatment, you will have follow-up tests or check-ups. Some tests that were done to diagnose or stage the cancer may be repeated to see how well the treatment is working. Decisions about whether to continue, change, or stop treatment may be based on the results of these tests.

Some of the tests will continue to be done from time to time after treatment has ended. The results of these tests can show if your condition has changed or if the cancer has recurred (come back).

## Treatment of Localized Transitional Cell Cancer of the Renal Pelvis and Ureter

Treatment of localized transitional cell cancer of the renal pelvis and ureter may include:

- surgery (nephroureterectomy or segmental resection of ureter)
- a clinical trial of fulguration
- a clinical trial of laser surgery
- a clinical trial of segmental resection of the renal pelvis
- a clinical trial of regional chemotherapy or regional immunotherapy

To learn more about these treatments, see the [Treatment Option Overview](#).

Use our [clinical trial search](#) to find NCI-supported cancer clinical trials that are accepting patients. You can search for trials based on the type of cancer, the age of the patient, and where the trials are being done. [General information](#) about clinical trials is also available.

## Treatment of Regional Transitional Cell Cancer of the Renal Pelvis and Ureter

Treatment of regional transitional cell cancer of the renal pelvis and ureter is usually done in a clinical trial.

To learn more about clinical trials, see the [Treatment Option Overview](#).

Use our [clinical trial search](#) to find NCI-supported cancer clinical trials that are accepting patients. You can search for trials based on the type of cancer, the age of the patient, and where the trials are being done. [General information](#) about clinical trials is also available.

## Treatment of Metastatic or Recurrent Transitional Cell Cancer of the Renal Pelvis and Ureter

Treatment of metastatic or recurrent transitional cell cancer of the renal pelvis and ureter is usually done in a clinical trial, which may include chemotherapy.

To learn more about clinical trials, see the [Treatment Option Overview](#).

Use our [clinical trial search](#) to find NCI-supported cancer clinical trials that are accepting patients. You can search for trials based on the type of cancer, the age of the patient, and where the trials are being done. [General information](#) about clinical trials is also available.

## To Learn More About Transitional Cell Cancer of the Renal Pelvis and Ureter

For more information from the National Cancer Institute about transitional cell cancer of the renal pelvis and ureter, see:

- [Kidney Cancer Home Page](#)
- [Tobacco](#) (includes help with quitting)

For general cancer information and other resources from the National Cancer Institute, visit:

- [About Cancer](#)
- [Cancer Staging](#)
- [Chemotherapy and You: Support for People With Cancer](#)
- [Radiation Therapy and You: Support for People With Cancer](#)
- [Coping with Cancer](#)
- [Questions to Ask Your Doctor about Cancer](#)
- [For Survivors, Caregivers, and Advocates](#)

## About This PDQ Summary

### About PDQ

Physician Data Query (PDQ) is the National Cancer Institute's (NCI's) comprehensive cancer information database. The PDQ database contains summaries of the latest published information on cancer prevention, detection, genetics, treatment, supportive care, and complementary and alternative medicine. Most summaries come in two versions. The health professional versions have detailed information written in technical language. The patient versions are written in easy-to-understand, nontechnical language. Both versions have cancer information that is accurate and up to date and most versions are also available in [Spanish](#).

PDQ is a service of the NCI. The NCI is part of the National Institutes of Health (NIH). NIH is the federal government's center of biomedical research. The PDQ summaries are based on an independent review of the medical literature. They are not policy statements of the NCI or the NIH.

## Purpose of This Summary

This PDQ cancer information summary has current information about the treatment of transitional cell cancer of the renal pelvis and ureter. It is meant to inform and help patients, families, and caregivers. It does not give formal guidelines or recommendations for making decisions about health care.

## Reviewers and Updates

Editorial Boards write the PDQ cancer information summaries and keep them up to date. These Boards are made up of experts in cancer treatment and other specialties related to cancer. The summaries are reviewed regularly and changes are made when there is new information. The date on each summary ("Updated") is the date of the most recent change.

The information in this patient summary was taken from the health professional version, which is reviewed regularly and updated as needed, by the [PDQ Adult Treatment Editorial Board](#).

## Clinical Trial Information

A clinical trial is a study to answer a scientific question, such as whether one treatment is better than another. Trials are based on past studies and what has been learned in the laboratory. Each trial answers certain scientific questions in order to find new and better ways to help cancer patients. During treatment clinical trials, information is collected about the effects of a new treatment and how well it works. If a clinical trial shows that a new treatment is better than one currently being used, the new treatment may become "standard." Patients may want to think about taking part in a clinical trial. Some clinical trials are open only to patients who have not started treatment.

Clinical trials can be found online at [NCI's website](#). For more information, call the [Cancer Information Service](#) (CIS), NCI's contact center, at 1-800-4-CANCER (1-800-422-6237).

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## Contact Us

More information about contacting us or receiving help with the Cancer.gov website can be found on our [Contact Us for Help](#) page. Questions can also be submitted to Cancer.gov through the website's [E-mail Us](#).

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